# ROHITH REDDY DEPA

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## EXPERIENCE

**University of Houston** — Oct 2023–Present
*Full Stack Developer* | Houston, TX

- Developed a modular, scalable UI using **Blazor** and **Razor Components** with **SignalR**, **form validation**, route guards, **state containers**, and **OnAfterRenderAsync** optimization for dynamic rendering.
- Built asynchronous, RESTful **ASP.NET Core Web APIs** with **CQRS** via **MediatR**, integrated **Swagger/OpenAPI**, centralized error handling via **Middleware**, and exposed **gRPC** endpoints for inter-service communication.
- Optimized **SQL Server** operations using **EF Core Code First** with **Redis caching**, improving response time by **50%**, utilizing complex LINQ queries, **eager/lazy loading**, and maintaining transactional integrity with **TransactionScope**.
- Deployed services to **Azure App Service** using **Docker**, integrated **SQL Server** via **Azure SQL**, managed static assets with **Blob Storage**, and implemented **Application Insights** and **Auto-scaling** for monitoring and reliability.

**OpenText** — Nov 2021–Aug 2023
*Software Engineer* | Hyderabad, India

- Developed a high-performance full-stack application using **Blazor** and **ASP.NET Core**, reducing **API response time by 35%** with **Redis cache** and improving UI load speed by **40%** via **lazy loading** and pre-rendering.
- Designed modular, responsive UI components using **Blazor Server**, with **SignalR**, **route guards**, and **component-level state management** for real-time data handling and seamless routing.
- Improved **SQL Server** performance by using **indexed views**, **partitioning**, and **query tuning**, increasing report speed by **40%**.
- Built a scalable **ETL pipeline** using **Azure Functions**, **Blob Storage**, and **Azure Service Bus**, reducing processing time by **60%**, while managing parallel batch execution with durable triggers.
- Designed CI/CD workflows using **GitHub Actions** with multistage **YAML**, deployed containerized .NET services via **Docker** and **Helm** on **Azure Kubernetes Service (AKS)**.

## TECHNICAL SKILLS

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|---|---|
| **Programming Languages** | C#, Java, Python, JavaScript/TypeScript, C/C++, Bash, Shell |
| **Web Development** | ASP.NET Core, Angular, React, Redux, Node.js, Blazor, Spring Boot, GraphQL, REST APIs, Razor Pages |
| **Databases** | SQL Server, PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, CockroachDB, Firebase, SQLite |
| **Cloud** | Azure, AWS, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins |
| **Data** | Apache Kafka, AWS Glue, Azure Data Factory, Pandas, NumPy, ETL pipelines, Redis caching |
| **CI/CD & Tools** | Git, Azure DevOps, Postman, SonarQube, Jira, Confluence, Visual Studio, VS Code, PyCharm, Eclipse |

## EDUCATION

**University of Houston** — May 2025
M.S. in Computer Science | GPA: 3.67/4.0
**Relevant Coursework:** Cloud Computing, Machine Learning, Artificial Intelligence, Database Management, Computer Architecture, Digital Image Processing.

## PROJECTS

**School Management Software**

- Architected a school management platform using **Spring Boot** and **React** on **AWS**, implementing **JWT/OAuth2** for secure authentication, **REST APIs** for seamless communication, and **MySQL** for reliable data storage.
- Implemented **Redux** for state management, **role-based access control (RBAC)** for security, and **Docker** for scalable deployment. Integrated **Razorpay** for online payments, processing **1,000+ monthly transactions**.

**ISP Messaging System**

- Engineered a scalable messaging system using **Apache Kafka** for real-time microservice communication.
- Integrated **OpenAI API** for query resolution, reducing customer support workload by **40%** through automated responses.
- Utilized **PostgreSQL** for persistent storage and **Docker** to containerize services, enabling seamless deployment.

**Traffic Asset Defect Detection for TxDOT**

- Designed an **AI-powered computer vision system** using **YOLOv5** with **transfer learning** and **OpenCV**-based preprocessing to detect traffic asset defects, improving detection accuracy by **25%**.
- Automated defect reporting pipeline using **Python scripting** and scheduled batch processing, reducing manual inspection and documentation time by **40%**.